

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Class : SY-IT	Batch : 00
Date of Assignment : 12-08-2026	Date/Time of Submission : 12-08-2026

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

```
import pandas as pd

data = {
    "name": ["Abhishek", "Pradeep", "Divyanshu"],
    "Roll No": [46, 30, 39],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)

print(df)
print("\nDivyanshu Singh S039")
```

```
      name  Roll No  qualified
0  Abhishek      46        True
1   Pradeep      30        False
2  Divyanshu      39        False
```

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5b. Write a python code to show statistical information on given data set.

```
import statistics
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print("\nDivyanshu Singh S039")
```

```
6
6.0
1
3.7416573867739413
14
```

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5c. Write a python code to create pandas series from dictionaries.

Example 1: sorted order.

```
import pandas as pd
dictionary = {'Pradeep': 30, 'Divyanshu': 39, 'Abhishek': 46}
series = pd.Series(dictionary)
print(series)

print("\nDivyanshu Singh S039")
```

```
Pradeep      30
Divyanshu    39
Abhishek     46
dtype: int64
```

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Example 2: index list (missing element is filled with NaN (Not a Number)).

```
import pandas as pd
dictionary = {'Divyanshu': 39, 'Pradeep': 30, 'Abhishek': 46}
series = pd.Series(dictionary, index=['Pradeep', 'Divyanshu', 'Abhishek', 'Shaan'])
print(series)
```

```
print("\nDivyanshu Singh S039")
```

```
Pradeep      30.0
Divyanshu    39.0
Abhishek     46.0
Shaan        NaN
dtype: float64
```

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5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

```
# importing pandas as pd
import pandas as pd

# dictionary of lists
dict = {'name':["Divyanshu", "Pradeep", "Abhishek", "Shaan"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score': [78, 80, 45, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])
print(df)
print("\nDivyanshu Singh S039")
```

```
      name degree score
True  Divyanshu   MBA   78
False   Pradeep   BCA   80
True   Abhishek M.Tech   45
False    Shaan   MBA   98
```

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